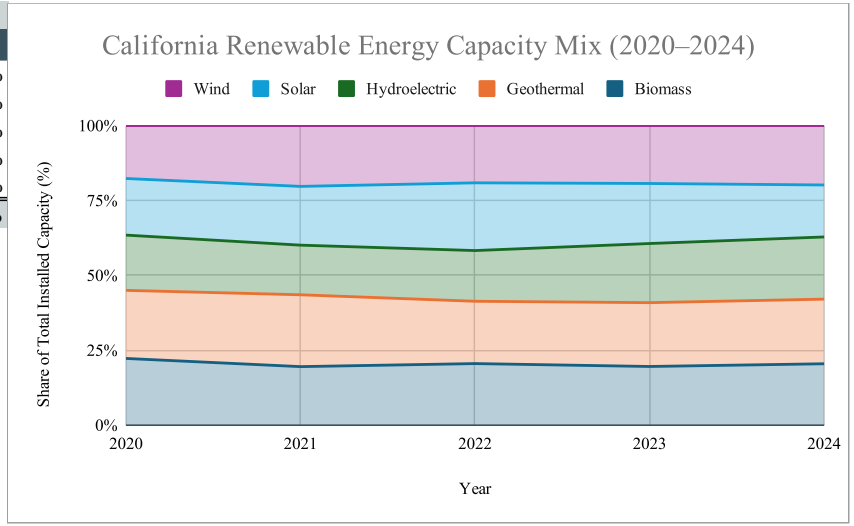
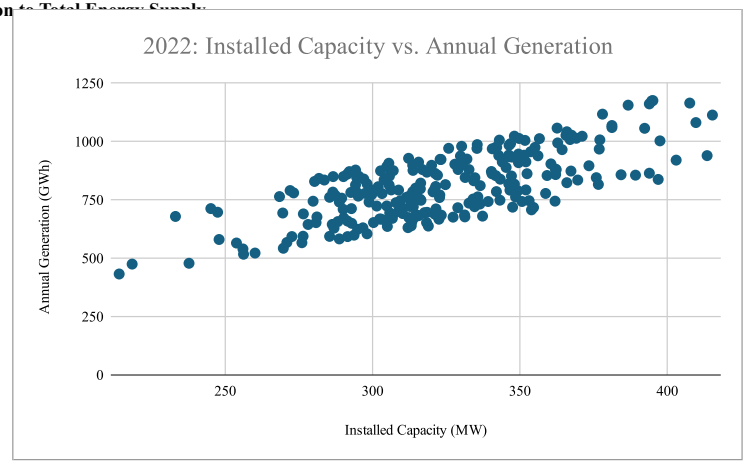
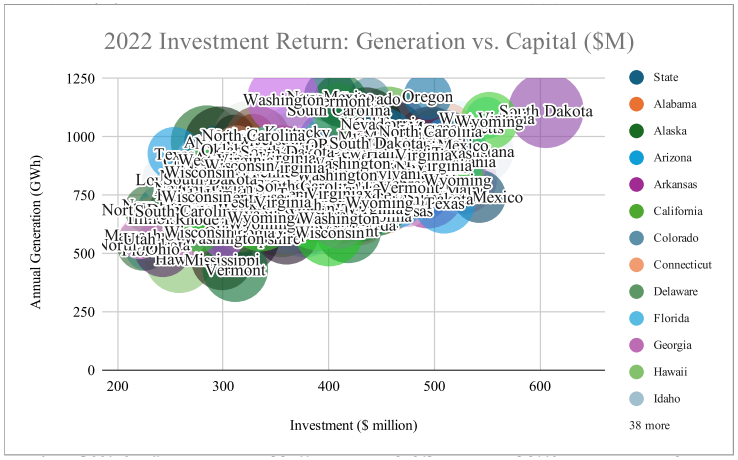
State	Year	Renewable Energy Type	Installed Capacity (MW)	Annual Generation (GWh)	Investment ($ million)	Number of Projects	% Contribution to Total Energy Supply
California	2020	Wind	217.8	581.3	196.4	7	13.2
California	2020	Solar	233.1	607.4	244.7	7	39.3
California	2020	Hydroelectric	227.6	556.8	319.8	8	11.1
California	2020	Geothermal	280.4	575.2	349.9	5	23.2
California	2020	Biomass	274.5	635.7	317	6	49.3
California	2021	Wind	304	685.4	354.3	10	33.7
California	2021	Solar	294	596.8	298.6	4	10.9
California	2021	Hydroelectric	248.3	712.1	229.4	4	33.4
California	2021	Geothermal	359.4	964.4	412.6	4	41.7
California	2021	Biomass	292.5	857.4	375.2	3	30.4
California	2022	Wind	322	856.2	346.4	8	29.3
California	2022	Solar	381.2	1060.5	456.2	12	6.3
California	2022	Hydroelectric	285.4	761.4	389.8	4	44
California	2022	Geothermal	350.7	743.5	414.2	12	17.1
California	2022	Biomass	346.2	819.2	372.9	8	19.6
California	2023	Wind	332.8	741.3	283.3	6	9.3
California	2023	Solar	345.3	850.7	368	11	26.4
California	2023	Hydroelectric	340.1	842.3	365.9	10	40.8
California	2023	Geothermal	367.1	741	344	10	44.9
California	2023	Biomass	337.1	694.6	490.2	9	10.5
California	2024	Wind	415.1	936.8	414.9	7	29.5
California	2024	Solar	363.5	1082	500.2	10	34.4
California	2024	Hydroelectric	434.8	1097.8	563.6	12	39.9
California	2024	Geothermal	451.9	1152.5	375.3	11	32.3
California	2024	Biomass	428.9	1060.7	352.9	12	32.7

SUM of Installe Renewable Energy Type						
Year	Biomass	Geothermal	Hydroelectric	Solar	Wind	Grand Total
2020	22.26%	22.73%	18.45%	18.90%	17.66%	100.00%
2021	19.52%	23.99%	16.57%	19.62%	20.29%	100.00%
2022	20.54%	20.81%	16.93%	22.62%	19.10%	100.00%
2023	19.57%	21.31%	19.75%	20.05%	19.32%	100.00%
2024	20.48%	21.58%	20.76%	17.36%	19.82%	100.00%
Grand Total	**20.39%**	**21.98%**	**18.66%**	**19.64%**	**19.33%**	**100.00%**



State	Year	Renewable En	Installed Capa	Annual Gener	Investment ($	Number of Pro	% Contribution to Total Energy Supply
Alabama	2022	Wind	268.4	764.6	338.2	3	27.6
Alabama	2022	Solar	306.9	688.7	454.6	6	44.7
Alabama	2022	Hydroelectric	292.8	712.9	306	3	29.3
Alabama	2022	Geothermal	341.8	854.2	410.3	4	3.1
Alabama	2022	Biomass	348.1	1023.6	510.9	8	44.3
Alaska	2022	Wind	269.7	543	230	3	27.2
Alaska	2022	Solar	294.2	814.5	429.9	5	13.1
Alaska	2022	Hydroelectric	291.4	860.8	356.8	3	33.9
Alaska	2022	Geothermal	321	794.7	443.5	5	42.8
Alaska	2022	Biomass	330.2	980.1	284.6	12	29.9
Arizona	2022	Wind	362.6	1058.1	439	6	13.2
Arizona	2022	Solar	233.1	679.4	245.6	8	11.8
Arizona	2022	Hydroelectric	336.4	732.6	495.4	4	28.3
Arizona	2022	Geothermal	352	913.6	298.1	10	21.9
Arizona	2022	Biomass	349.7	1013.6	330	4	2.8
Arkansas	2022	Wind	348.5	762.8	493.8	12	2.4
Arkansas	2022	Solar	318.9	638.1	379.3	10	10.5
Arkansas	2022	Hydroelectric	297.4	778.3	442.2	3	6.7
Arkansas	2022	Geothermal	337.3	680.8	371.9	5	17.8
Arkansas	2022	Biomass	328.9	716.8	436.6	9	46.9
California							29.3
California							6.3
California							44
California							17.1
California							19.6
Colorado							47.8
Colorado							42.2
Colorado							33.3
Colorado							6.2
Colorado							5.2
Connecticut							8.9
Connecticut							4.2
Connecticut							40.1
Connecticut							10.6
Connecticut							29.3
Delaware							20.8
Delaware							19.8
Delaware							16.6
Delaware							32.6
Delaware	2022	Biomass	315.6	912.2	328.9	7	17.6
Florida	2022	Wind	294.7	781.5	388.2	8	32.6
Florida	2022	Solar	299.7	773	257.7	5	33.7
Florida	2022	Hydroelectric	396.9	838.1	460.9	8	34.7
Florida	2022	Geothermal	256.2	517.4	227.3	5	25.2
Florida	2022	Biomass	272	790.4	268.4	11	48.2
Georgia	2022	Wind	318.3	653	367.1	9	6.8
Georgia	2022	Solar	365.6	1026.9	465.8	10	16.5
Georgia	2022	Hydroelectric	301.9	806.9	409.9	10	16.3
Georgia	2022	Geothermal	315.6	682.7	293.8	8	16
Georgia	2022	Biomass	336.5	811.4	489.4	8	36.9
Hawaii	2022	Wind	237.7	478.7	258.4	11	40.4
Hawaii	2022	Solar	313.1	639.3	355.5	12	37.4
Hawaii	2022	Hydroelectric	325.8	971.5	363.4	9	22.2
Hawaii	2022	Geothermal	330.5	901.9	405.1	10	19.1





Hawaii	2022	Biomass	346.2	942.6	436.1	5	33.4
Idaho	2022	Wind	362.8	994.6	361.2	9	42.9
Idaho	2022	Solar	314.7	687	433.7	9	30.6
Idaho	2022	Hydroelectric	305.5	907.5	366.6	12	39.5
Idaho	2022	Geothermal	305	636.6	362.9	8	9.6
Idaho	2022	Biomass	365.9	824.1	425.9	7	13.2
Illinois	2022	Wind	346.6	792.1	299.4	9	27
Illinois	2022	Solar	343.3	839.1	489.4	10	41.8
Illinois	2022	Hydroelectric	335.5	987.4	324.1	7	38.6
Illinois	2022	Geothermal	314.6	781.6	364.9	8	45.2
Illinois	2022	Biomass	280.8	652.9	232.4	5	41.6
Indiana	2022	Wind	318.4	869.2	392.7	12	14.9
Indiana	2022	Solar	384.3	858.1	368.3	11	48.2
Indiana	2022	Hydroelectric	313.8	721.2	312.1	12	5.2
Indiana	2022	Geothermal	327.3	676.4	272.8	9	26.6
Indiana	2022	Biomass	321.2	863.9	313.7	9	33
Iowa	2022	Wind	367.5	1027.6	483.1	4	7.5
Iowa	2022	Solar	278.1	645	303.4	11	12.9
Iowa	2022	Hydroelectric	312	631.5	377.4	11	7.3
Iowa	2022	Geothermal	286.3	645.2	357.7	11	2.1
Iowa	2022	Biomass	295.1	766.5	403.5	5	31
Kansas	2022	Wind	331.2	685.8	475.8	4	41.2
Kansas	2022	Solar	376.6	816.7	526.4	11	11.4
Kansas	2022	Hydroelectric	341.6	969	393.5	4	41.2
Kansas	2022	Geothermal	313.6	785.2	443.7	7	38
Kansas	2022	Biomass	312.3	674.5	400.9	5	16.2
Kentucky	2022	Wind	331.3	677.6	366.6	8	13.3
Kentucky	2022	Solar	320.5	784.9	386	9	33
Kentucky	2022	Hydroelectric	316.4	748.8	275.1	7	49.7
Kentucky	2022	Geothermal	403	920.8	500.3	9	15.2
Kentucky	2022	Biomass	371.1	1023.3	370.1	4	5.7
Louisiana	2022	Wind	362.1	858.9	313.1	9	13.9
Louisiana	2022	Solar	286.6	850.2	421.5	7	26.2
Louisiana	2022	Hydroelectric	335.4	971.1	495.2	12	44.6
Louisiana	2022	Geothermal	413.5	940.1	544	10	6.2
Louisiana	2022	Biomass	298.1	819.1	248.5	7	35.8
Maine	2022	Wind	303.2	777.5	296.1	6	13.7
Maine	2022	Solar	358.7	778	530.1	5	34.5
Maine	2022	Hydroelectric	329	880.6	332.6	11	4.5
Maine	2022	Geothermal	313.7	876.9	282	9	23.2
Maine	2022	Biomass	314.6	800.7	270.8	7	41.3
Maryland	2022	Wind	294.6	846.2	362.1	11	5.1
Maryland	2022	Solar	300.3	788.8	293.4	9	41.1
Maryland	2022	Hydroelectric	343	1006.6	324	11	37.9
Maryland	2022	Geothermal	369.6	835.5	372.5	10	48.2
Maryland	2022	Biomass	352.4	862.6	307.1	10	6
Massachusetts	2022	Wind	316.9	880.4	338.4	9	25.7
Massachusetts	2022	Solar	365.9	1042.4	520.4	7	36.7
Massachusetts	2022	Hydroelectric	247.9	580.6	232.1	7	16.4
Massachusetts	2022	Geothermal	360.5	904.7	426.1	12	21.9
Massachusetts	2022	Biomass	286.9	629.4	420.5	5	34.4
Michigan	2022	Wind	393.9	864.7	339	12	13.1
Michigan	2022	Solar	305.6	813.9	370.8	11	36.5
Michigan	2022	Hydroelectric	397.5	1003.4	499.3	6	27.5
Michigan	2022	Geothermal	342.8	940.5	379.4	7	7.8

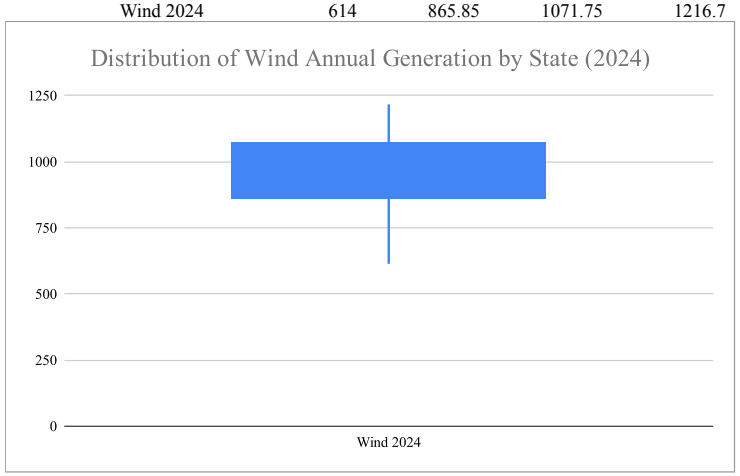
Michigan	2022	Biomass	304.8	893.1	446.1	11	17.6
Minnesota	2022	Wind	351.4	941.9	492.4	11	32.5
Minnesota	2022	Solar	335.7	763.4	456.8	12	24.7
Minnesota	2022	Hydroelectric	347.5	719.2	292.2	7	11.6
Minnesota	2022	Geothermal	377.1	1007.6	441.8	9	21.2
Minnesota	2022	Biomass	313.6	898.8	288.7	7	28.7
Mississippi	2022	Wind	218.4	475.2	298.9	9	13.9
Mississippi	2022	Solar	321.5	686.6	373.9	4	46.8
Mississippi	2022	Hydroelectric	334.3	832.5	370	6	19.8
Mississippi	2022	Geothermal	314.1	674.4	385.7	8	36
Mississippi	2022	Biomass	283.6	835.5	275.5	8	13.5
Missouri	2022	Wind	290.1	850.2	402.1	7	25.2
Missouri	2022	Solar	369.1	1014.5	497.9	6	36.3
Missouri	2022	Hydroelectric	280.3	829.4	406.9	5	36.4
Missouri	2022	Geothermal	305.8	694.7	326.8	12	12
Missouri	2022	Biomass	356.6	1012.9	460.3	10	10.6
Montana	2022	Wind	353.1	746.8	308.9	5	13.6
Montana	2022	Solar	310.2	691.8	360.2	5	47.1
Montana	2022	Hydroelectric	322.5	667.6	400.4	5	19.7
Montana	2022	Geothermal	341.8	975.6	495.8	12	20.2
Montana	2022	Biomass	389.2	856.4	424	7	2.8
Nebraska	2022	Wind	351.3	922.7	449.4	9	48.7
Nebraska	2022	Solar	346.8	802	478.1	9	27
Nebraska	2022	Hydroelectric	281.8	842.5	359.5	9	33.1
Nebraska	2022	Geothermal	291.5	659.5	404.4	6	12.4
Nebraska	2022	Biomass	288.7	741.2	344.4	7	39.7
Nevada	2022	Wind	294.9	624.4	439.5	3	2.5
Nevada	2022	Solar	322.9	924.2	475.6	6	17.4
Nevada	2022	Hydroelectric	355.1	975.4	297.2	12	13.7
Nevada	2022	Geothermal	351.7	1005.4	310.3	5	34
Nevada	2022	Biomass	392.3	1057.2	435.1	12	45.7
New Hampshire	2022	Wind	346.3	935.2	453.2	6	20.8
New Hampshire	2022	Solar	272.6	593.2	359.9	10	23.3
New Hampshire	2022	Hydroelectric	290	710	256.6	7	15.1
New Hampshire	2022	Geothermal	276	566.8	320.6	11	8.3
New Hampshire	2022	Biomass	298.8	741.1	433.8	5	30.9
New Jersey	2022	Wind	286.5	783.7	271.6	10	22.1
New Jersey	2022	Solar	298.1	605.1	321.3	6	49.6
New Jersey	2022	Hydroelectric	247.4	697.5	292.7	4	42.1
New Jersey	2022	Geothermal	346.8	991.5	332.8	10	29.5
New Jersey	2022	Biomass	307	875.7	412.1	11	23.8
New Mexico	2022	Wind	394.7	1174.6	401.6	7	35
New Mexico	2022	Solar	361.9	745	542.5	7	44.1
New Mexico	2022	Hydroelectric	364.3	965.8	509.8	11	43.9
New Mexico	2022	Geothermal	321.4	807.8	388.9	9	42.9
New Mexico	2022	Biomass	306.7	671.3	269.8	12	21.5
New York	2022	Wind	362	882.6	295.1	11	10.5
New York	2022	Solar	304.8	724.3	387.2	11	41.8
New York	2022	Hydroelectric	306.3	793.1	288	10	26.1
New York	2022	Geothermal	340.3	872	497.3	10	32.4
New York	2022	Biomass	308.8	701.2	333.6	7	13.9
North Carolina	2022	Wind	366.3	1026.3	496.1	6	29.8
North Carolina	2022	Solar	353.9	707.8	372.9	8	36.1
North Carolina	2022	Hydroelectric	367	1009.1	328.5	5	46.7
North Carolina	2022	Geothermal	322.8	771.5	347	10	7.1

North Carolina	2022	Biomass	349.5	909.9	355.9	10	47.7
North Dakota	2022	Wind	287.3	770	364.4	3	46.4
North Dakota	2022	Solar	307.9	738.3	313.2	6	49
North Dakota	2022	Hydroelectric	276.5	690.3	229.2	6	3.4
North Dakota	2022	Geothermal	281.1	677.3	298.6	3	37.7
North Dakota	2022	Biomass	256	540	224.2	7	9
Ohio	2022	Wind	269.5	694.1	325.6	9	4.6
Ohio	2022	Solar	260.1	522.5	242.7	8	26.5
Ohio	2022	Hydroelectric	353.6	956.6	493.4	9	29.9
Ohio	2022	Geothermal	339.2	743.4	324.7	11	45.1
Ohio	2022	Biomass	273.2	780.4	273.5	3	19.6
Oklahoma	2022	Wind	349.5	946.8	312.5	6	30.7
Oklahoma	2022	Solar	292.6	782.2	322.7	11	10.6
Oklahoma	2022	Hydroelectric	303.9	836.7	354.8	5	4.2
Oklahoma	2022	Geothermal	310.2	714.8	345.9	6	5.9
Oklahoma	2022	Biomass	285.4	594	315.4	5	20
Oregon	2022	Wind	346.3	984.8	451.2	10	17.6
Oregon	2022	Solar	395.1	1176.1	493.6	6	10.8
Oregon	2022	Hydroelectric	376.9	968.2	402.4	11	49.5
Oregon	2022	Geothermal	300.1	792.4	367.8	7	31.5
Oregon	2022	Biomass	318.5	697	381.6	8	17
Pennsylvania	2022	Wind	290.2	673	311.9	3	34.5
Pennsylvania	2022	Solar	373.4	896.7	515.6	12	22.2
Pennsylvania	2022	Hydroelectric	343.2	749.1	421.1	10	39.4
Pennsylvania	2022	Geothermal	313	747.7	372.5	11	9.1
Pennsylvania	2022	Biomass	375.9	845.6	451.6	7	24.6
Rhode Island	2022	Wind	334.1	754	279.5	4	26.2
Rhode Island	2022	Solar	292.9	649.9	296.9	3	36
Rhode Island	2022	Hydroelectric	313.3	656.3	411.2	7	29.4
Rhode Island	2022	Geothermal	340.6	969.5	432.3	6	10
Rhode Island	2022	Biomass	312.3	765.1	415.8	5	7.5
South Carolina	2022	Wind	378	1117.9	409.3	7	10.3
South Carolina	2022	Solar	331.4	846.4	345.6	11	4.6
South Carolina	2022	Hydroelectric	308.8	792.4	379.3	12	32.1
South Carolina	2022	Geothermal	304.3	662.6	429.5	7	3.4
South Carolina	2022	Biomass	323.3	684.9	265.2	11	22.1
South Dakota	2022	Wind	415.3	1114	605.6	13	37.2
South Dakota	2022	Solar	316.3	822.3	287.6	5	27.4
South Dakota	2022	Hydroelectric	356.1	939.4	360.7	5	49.9
South Dakota	2022	Geothermal	332.7	736.5	492.5	5	9.7
South Dakota	2022	Biomass	342.1	975.9	444.8	5	19
Tennessee	2022	Wind	289.5	757.8	383.9	5	12.2
Tennessee	2022	Solar	321	690.3	444.6	6	36.9
Tennessee	2022	Hydroelectric	349.3	789.1	464	5	39.8
Tennessee	2022	Geothermal	347.1	852.9	357.5	9	16.5
Tennessee	2022	Biomass	279.8	744.7	331.6	8	6.5
Texas	2022	Wind	311.5	761.3	434.7	9	18.4
Texas	2022	Solar	354.6	717.9	509.2	9	4.7
Texas	2022	Hydroelectric	316.1	753.4	460.9	11	2.7
Texas	2022	Geothermal	312.2	928.6	253.5	7	40.8
Texas	2022	Biomass	347.3	933.8	516.2	12	8.9
Utah	2022	Wind	340.5	862.3	299.6	10	1.1
Utah	2022	Solar	367.3	874	509.4	8	28.9
Utah	2022	Hydroelectric	321.5	711.7	377.9	9	32.1
Utah	2022	Geothermal	288.7	583.1	259.4	7	10.1

Utah	2022	Biomass	253.8	565.5	220.6	5	10.7
Vermont	2022	Wind	312.7	722.1	444	11	44.5
Vermont	2022	Solar	386.7	1156.4	410.8	7	2.6
Vermont	2022	Hydroelectric	342	785.9	475.9	9	43.8
Vermont	2022	Geothermal	293.7	599.2	418.5	10	9.9
Vermont	2022	Biomass	214	432.7	311.4	10	27.6
Virginia	2022	Wind	302.4	668.6	389.4	12	31.2
Virginia	2022	Solar	362.3	880.2	508.4	8	14.7
Virginia	2022	Hydroelectric	322.7	760.5	405.4	5	46
Virginia	2022	Geothermal	317.6	695.5	443.2	6	5.3
Virginia	2022	Biomass	332	925.4	489.5	4	45.9
Washington	2022	Wind	345.3	889.5	419.9	8	44.8
Washington	2022	Solar	300.3	653.2	409.7	9	22.2
Washington	2022	Hydroelectric	270.9	567.8	300.3	5	9.6
Washington	2022	Geothermal	297	839.8	407.5	10	15.2
Washington	2022	Biomass	393.9	1161.9	356.9	12	22.1
West Virginia	2022	Wind	409.7	1081.8	549.4	5	31
West Virginia	2022	Solar	344	912.8	342.1	9	17.4
West Virginia	2022	Hydroelectric	327.9	903.3	302.3	6	16.1
West Virginia	2022	Geothermal	292.1	871.6	350.4	8	28
West Virginia	2022	Biomass	301.4	723.9	337.8	11	4
Wisconsin	2022	Wind	332.7	880.3	315	10	29.8
Wisconsin	2022	Solar	276.4	594.6	277.2	5	3.1
Wisconsin	2022	Hydroelectric	309.4	747	274.4	10	10.5
Wisconsin	2022	Geothermal	295.3	851.7	277.5	6	21.2
Wisconsin	2022	Biomass	291.5	593	400.3	11	14.8
Wyoming	2022	Wind	296.7	630.7	336.1	5	3.3
Wyoming	2022	Solar	348.1	817.9	521.7	7	37.7
Wyoming	2022	Hydroelectric	381.2	1069.2	552.1	8	26.7
Wyoming	2022	Geothermal	288.2	657.4	335.7	11	11.3
Wyoming	2022	Biomass	334.6	721.3	447.7	4	28

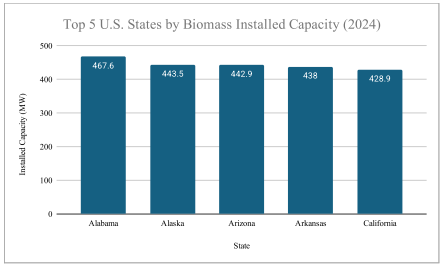
State	Year	Renewable En	Installed Capa	Annual Gene	Investment ($	Number of Pro	% Contribution to Total Ener
Alabama	2024	Wind	415.4	864.6	333.8	8	40.1
Alaska	2024	Wind	384.1	817.4	348.1	9	35.2
Arizona	2024	Wind	323.1	678.4	313.5	11	40.5
Arkansas	2024	Wind	409.3	1190.9	573.1	8	20.5
California	2024	Wind	415.1	936.8	414.9	7	29.5
Colorado	2024	Wind	381.5	874.7	566.9	9	24.8
Connecticut	2024	Wind	406.8	879.5	329	13	5.6
Delaware	2024	Wind	348.6	1007	426.1	12	49.2
Florida	2024	Wind	398.3	887.5	387.4	10	33.9
Georgia	2024	Wind	370.3	887.8	364.7	5	24.7
Hawaii	2024	Wind	442.9	967.8	540.2	6	21.8
Idaho	2024	Wind	375.1	767.2	469.8	6	31.1
Illinois	2024	Wind	423	915	407.6	9	41.9
Indiana	2024	Wind	339.8	838.2	290.1	6	33.2
Iowa	2024	Wind	304.1	614	400	12	44.1
Kansas	2024	Wind	307.7	869.6	396.2	10	41.2
Kentucky	2024	Wind	364.4	1005.8	464	7	38.1
Louisiana	2024	Wind	330.6	912.5	431.6	12	11.7
Maine	2024	Wind	292.9	861.4	270.2	11	31.6
Maryland	2024	Wind	405.5	1148.3	438.9	5	46.8
Massachusetts	2024	Wind	448.7	991.3	625.5	5	27.5
Michigan	2024	Wind	420.2	1158.3	433.8	10	20.3
Minnesota	2024	Wind	410.4	1213.2	481	5	48.5
Mississippi	2024	Wind	379.7	969.1	317.2	8	22.8
Missouri	2024	Wind	388.2	1095.1	320.8	6	47.6
Montana	2024	Wind	386.7	1005.2	519.5	10	47.1
Nebraska	2024	Wind	391.4	1167.3	367.4	8	3.8
Nevada	2024	Wind	319.2	682	426	12	12.3
New Hampshir	2024	Wind	402.9	988	326.9	7	10.1
New Jersey	2024	Wind	406	1181.8	556.2	10	19.5
New Mexico	2024	Wind	415	839.4	531.6	7	28.4
New York	2024	Wind	390	987.5	532.9	4	10.8
North Carolina	2024	Wind	435.2	1057.2	642.5	12	46.9
North Dakota	2024	Wind	365.6	885.7	497.3	4	39.5
Ohio	2024	Wind	412.5	988.8	391.3	8	7.6
Oklahoma	2024	Wind	332.8	934.3	334.8	5	31.4
Oregon	2024	Wind	387.1	1046.2	565.2	10	22.6
Pennsylvania	2024	Wind	435.3	1091.5	593.3	6	28.9
Rhode Island	2024	Wind	414.8	1156	465.9	6	46.1
South Carolina	2024	Wind	442.4	900.6	603.1	6	42.8
South Dakota	2024	Wind	387.1	780.1	531	7	18.6
Tennessee	2024	Wind	442.5	1216.7	408.4	5	6.4
Texas	2024	Wind	392.9	1169.1	415.2	11	21.9
Utah	2024	Wind	362.8	882	417.7	5	13.8
Vermont	2024	Wind	388.4	1142.4	541.7	6	27.2
Virginia	2024	Wind	327.5	694	285.5	6	2.5
Washington	2024	Wind	373.5	1038.4	496.4	4	1.2
West Virginia	2024	Wind	343.1	852.5	475	5	6.7
Wisconsin	2024	Wind	323.8	753.2	436.3	4	39.3
Wyoming	2024	Wind	362.5	1076.6	339.9	9	1.7

Category	Low	Open	Close	High
Wind 2024	614	865.85	1071.75	1216.7



Copy the top 5 rows (State + Installed Capacity) into the Solutions sheet

State	Year	Renewable En	Installed Capacity (MW)
Alabama	2024	Biomass	467.6
Alaska	2024	Biomass	443.5
Arizona	2024	Biomass	442.9
Arkansas	2024	Biomass	438
California	2024	Biomass	428.9



1. Which state had the highest installed capacity for Biomass in 2024?
Alabama had the highest installed capacity for Biomass in 2024 at 467.6 MW.

2. What's the difference in MW between the top and bottom state in your chart?
Top State (Alabama): 467.6
MW Bottom State in Chart (California): 428.9 MW
Difference: 467.6 MW - 428.9 MW = 38.7 MW

3. ***What could this visual tell you about state-level investments in Biomass energy?***
High Level of Parity Among Leaders: The top 5 states are tightly grouped together, separated by only 38.7 MW across the entire group. This suggests that leading states are making substantial, comparable investments in Biomass infrastructure rather than one single state dominating the market.

Geographic Diversity: Biomass investment is spread across diverse regions (from the Southeast like Alabama/Arkansas to Alaska and the West Coast like California), showing that Biomass is viable across different climate zones and industrial profiles.

4. What factors might explain why some states have significantly higher capacity than others?

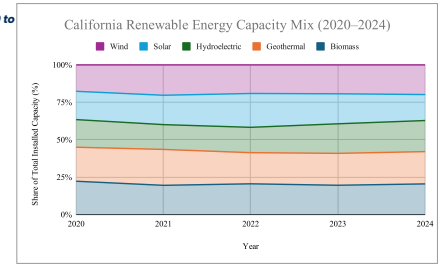
Abundance of Organic Feedstock & Resources: States with large forestry industries (e.g., Alabama, Arkansas), extensive agriculture, or wood waste have cheap, ready access to biomass materials (wood scraps, agricultural waste).

State Policy & Incentives: Renewable Portfolio Standards (RPS), state tax credits, or carbon reduction mandates encourage heavier capital investment in bioenergy facilities.

Grid Infrastructure & Existing Facilities: States with existing paper/pulp plants or bio-processing facilities can retrofit infrastructure for bioenergy generation more cost-effectively than building from scratch.

Land Availability & Population Size: Larger states have more space to establish biomass processing plants and manage supply chain logistics for organic waste collection.

PART TWO: In a 100% stacked area chart of California's renewable capacity from 2020 to what does it mean if one energy type's band is narrowing while another is widening?



What Narrowing vs. Widening Bands Mean
In a 100% stacked area chart, the vertical axis always represents 100% of the total capacity for that year, regardless of how much total capacity (in MW) grew in raw numbers.

A Widening Band: Means that specific energy type is increasing its share of the total portfolio relative to the other sources. It is growing at a faster rate than the rest of the overall mix.

A Narrowing Band: Means that energy type is losing relative share of the overall mix.

Crucial Key Insight: Relative Share vs. Absolute Growth
A narrowing band does NOT necessarily mean capacity is declining in raw megawatts.
For example, looking at California's data from 2020 to 2024:
Solar's band narrows slightly ***(falling from 18.90% of the mix in 2020 down to 17.36% in 2024)***. However, Solar's actual capacity increased from **233.1 MW to 363.5 MW** in that same timeframe. Because Hydroelectric **(18.45% => 20.76%)** and Wind **(17.66% => 19.82%)** grew at a slightly faster proportional rate during these years, their bands widened, causing Solar's relative slice of the 100% pie to contract.
In short:
Widening = Outpacing the portfolio's average growth rate.
Narrowing = Lagging behind the portfolio's average growth rate (even if its actual capacity is still rising)

Key Analytical Takeaways for the Policy Advisory Team
Does Installed Capacity correlate with Annual Generation?

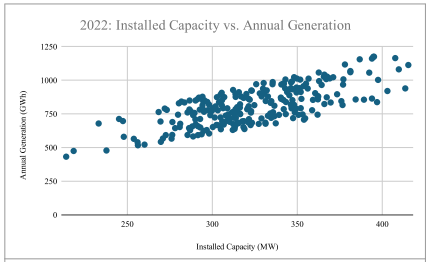
Yes, strongly. The scatter plot demonstrates a strong positive linear correlation (r = 0.736). As installed megawatt capacity increases, annual gigawatt-hour output increases predictably.

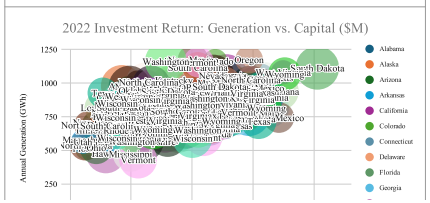
Where have Investments ($ million) had the greatest return?

Data points in the upper-left region of the bubble chart represent high efficiency/return (high generation output for lower total dollar investment). Conversely, bubbles in the lower-right represent lower capital efficiency.

How does Number of Projects factor in?

Larger bubbles (more projects) spread across lower individual generation outputs suggest fragmented, smaller-scale investments. Smaller, higher-positioned bubbles indicate consolidated, large-scale utility projects that yield significantly higher annual generation per dollar invested.

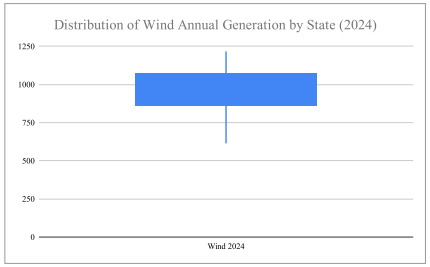




Copy the top 5 rows (State + Installed Capacity) into the Solutions sheet



What is the median annual generation?
968.45 GWh or 968.5 GWh
The median is the exact midpoint
(50th percentile) of state wind generation
-half of the U.S. states generated more
than this amount in 2024, and half generated less.
What is the interquartile range (IQR)?
226.75 GWh
Which (if any) states appear as outliers on the high end,
and what are their approximate generation values?
There are no high-end outliers in this 2024 dataset
without those policy drivers.



Grid Infrastructure: States with modernized grid capacity capable of integrating variable renewables can support far higher generation levels without severe curtailment.

3. Implications for Energy Equity
From a policy standpoint, a wide spread signals an uneven distribution
of economic and environmental benefits:

Economic Development: Leading states capture a disproportionate share of green jobs, landowner lease payments, and local tax revenues generated by wind farms.

Decarbonization Disparity: States at the bottom end of the IQR remain more reliant on fossil fuels, leading to localized air quality and health disparities compared to high-generation states.

A large IQR relative to the median signals that national wind generation is highly stratified, indicating a need for targeted federal support, regional grid integration, or tailored policy interventions to help lagging states build out their renewable capacity.

1. Scatter Chart
Question Asked: What is the relationship between Installed Capacity (MW) and Annual Generation (GWh)?

Why it was the best choice:
A scatter plot is specifically designed to plot two continuous numerical variables against each other on a Cartesian plane (X and Y). It is the industry-standard visualization for:

Evaluating correlation (whether generation increases proportionally with capacity).

Identifying the shape of the relationship (linear vs. non-linear).

Detecting clusters or unusual variance (states generating far above or below their capacity limit).

2. Bubble Chart
Question Asked: How do states compare in return on investment (ROI) across generation output, total capital expenditure, and project scale?

Why it was the best choice:
A standard 2D chart can only plot two variables (X and Y). The analytical question required comparing four distinct data dimensions simultaneously:

Horizontal (X): Investment ($ million)

Vertical (Y): Annual Generation (GWh)

Bubble Size: Number of Projects (Scale/Fragmentation)

Color (Series): Renewable Energy Type (Categorical group)

A bubble chart expands on a scatter plot by encoding a 3rd numerical metric into area/size and a 4th metric into color, allowing you to instantly assess capital efficiency (top-left positions) and project consolidation (bubble size) in a single visual.

3. Box Plot (Candlestick Chart)
Question Asked: What is the distribution, central tendency, spread, and outlier profile of Wind energy generation across all U.S. states?

Why it was the best choice:
Bar charts or pie charts show aggregate sums or averages, but they completely hide data dispersion and skewness. A box plot is the ideal statistical visualization because it condenses an entire 50-state distribution into a clean 5-number summary:

Central Tendency: Pinpoints the median (50th percentile), avoiding the distortion caused by extreme values in standard averages.

Spread & Skew: Visualizes the Interquartile Range (IQR, middle 50%) and overall range via whiskers.

Outlier Detection: Highlights extreme observations that fall beyond 1.5×IQR, making it effortless to identify regional disparities and runaway state outputs.